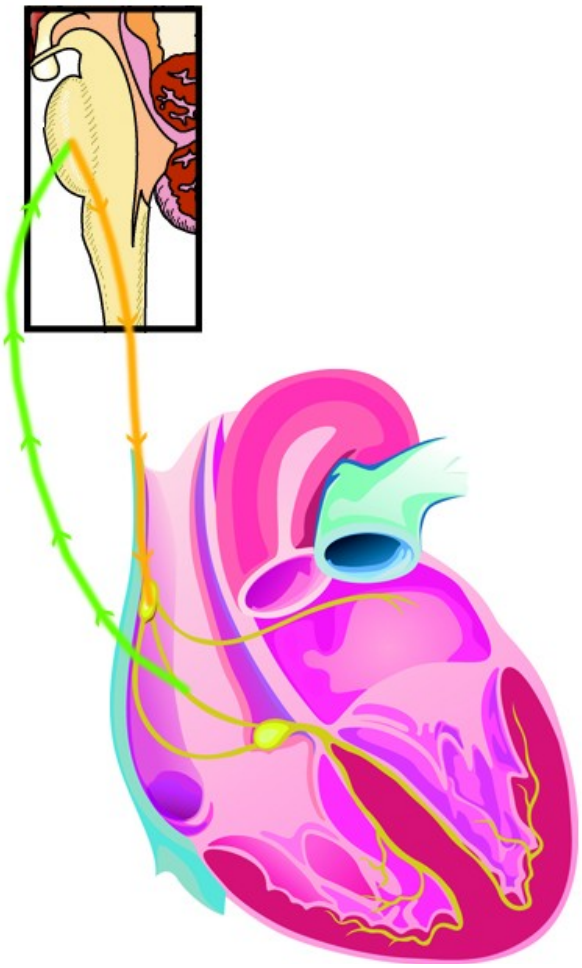
When venous return to the heart increases, the heart rate will increase by as much as 75% to compensate, thereby increasing the cardiac output to maintain forward blood flow. Two factors contribute to this effect: the first is the direct effect of stretching of the sinus node, which stimulates an increase in it's own firing rate, the second, and most significant, is due to specific stretch receptors within the atria that elicit the Bainbridge reflex.

The stretch receptors within the atria transmit afferent signals to the medulla. The medulla then responds via vagal and sympathetic fibers to increase heart rate and contractility. Thus, the Bainbridge reflex increases heart rate and contractility in response to atrial dilation from increased venous return to prevent the congestion of blood in the atria and venous circulation.



Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 208-209, 229-230.

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